PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review
AUTHORS	Zhao, Yueqiu; Chen, Licong; Zheng, Xiaoqing; Shi, Yulan

VERSION 1 – REVIEW

REVIEWER	Fayh, Ana Universidade Federal do Rio Grande do Norte
REVIEW RETURNED	04-Apr-2022

GENERAL COMMENTS	Thank you for your confidence and the opportunity to review this paper. The scope of this article is unpublished and has relevance to the clinical area related to breast cancer treatment. However, I believe that the aim of this work should not be answered through a systematic review methodology. Due to the different objectives listed and the discrepancy of the studies included in the article, conducting a scoping review would be most appropriate. The general purpose for conducting scoping reviews is to identify and map the available evidence. Also, scoping reviews are useful for examining emerging evidence when it is still unclear what other, more specific questions can be posed and valuably addressed by a more precise systematic review. Based on the above consideration, I believed that the manuscript deserves careful revision before being resubmitted for review. Lastly, I did not find the review registration number on the appropriate platform.

REVIEWER	Rahman, M.Mizanur National Institute of Cancer Research and Hospital, Surgical Oncology
REVIEW RETURNED	25-Apr-2022

GENERAL COMMENTS	Quality of life in breast cancer patients with neoadjuvant chemotherapy: a systematic review comments after review: P 3 Lines 8,9 – language to be reorganized Rationality of the study not clear in introduction P8 Neoadjuvant vs adjuvant chemotherapy : to be cleared P9 Rezapour et al.28 - Insomnia and diarrhea were better in the PG group. - Any better language? P 11

	Support from family or community; does it have any impact on NAC and QOL? P12 Line 54,55 to explain p 13. Conclusion- Is there any message or any recommendation for upfront treatment after NAC ? Apart from support that already mentioned, other factors like age, educational status, socioeconomic status etc have any influence on QOL? Now a days as a part of pre operative preparation some interventions are made like inserting marker inside tumor, shared decision making etc, does they have any effect on QOL? Remission of clinical parameters like shrinkage of tumor size, is there any reflection on post NAC QOL? Loss of hair makes a serious impact on QOL , does it have any effect on the type of surgery? Finally pre operative QOL improvement or reduction does it carry any impact on decision making for the type of surgery? Is there any evidence of pre operative vs post operative QOL mentioning for direction of type of surgery like mastectomy, traditional bcs, onciplastic surgery or breast reconstruction?

VERSION 1 – AUTHOR RESPONSE

To the reviewer 1:

1.From the beginning of the study design, we primarily followed the Joanna Briggs Institute guidelines on systematic reviews and we used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. We accepted your suggestion and have added a statement in the methods section

2.We had registered in PROSPERO with ID 303880 and we were told that more than 3 months waiting time needed for the applicant from outside the UK. Continue working was suggested. We have searched major electronic databases to make sure that there was no recently published systematic review on the same topic prior to registration. We identified the topic with QOL and we developed a focused, answerable question. The question is the influence factors of QOL and is specified in detail in the review.

3.Our research questions of systematic review are clearer and can be PICOS（Population is breast cancer patients, intervention is neoadjuvant chemotherapy, comparators are modalities other than neoadjuvant chemotherapy, outcomes are quality of life）, whereas the research questions of scoping review are broader and PCC (Population, Concept and Context). On synthesis of findings we found that the QOL are mainly reflected in physical symptoms, psychological effects and support from society or family. The QOL could be improved with appropriate interventions to chronotype, QOL before neoadjuvant chemotherapy and depression.

4.Our review matches the chart as the attachment.

5. Systematic review study on the QOL of pancreatic cancer, prophylactic mastectomy and cancer-related lower-extremity lymphedema patients were also published in the recent years (ref. 1,2,3).

6. The cohort study, RCT and cross-sectional studies were included in our study and similar discrepancy of the studies was used in the published papers of other system reviews (ref 4, 5).

To the reviewer 2:

Thank you very much for your suggestion. We have added supplements in the discussion section as the following:
(1)The initial stages of treatment and the months following the end of treatment can be difficult times for patients, both physically and emotionally, and BC patients are vulnerable to maladjustment and decreased QOL during these periods.
(2)A transient increase in physical symptoms and a decrease in QOL were seen after neoadjuvant therapy, and in those patients who underwent surgery, most domains returned back by several months.
(3)Past studies reported that women who are malnourished have a serious negative impact on QOL, and nutritional intervention can improve the symptoms of nausea, vomiting, and loss of appetite in BC patients, and exercise can reduce fatigue and improve the physical quality of BC patients. Good nutritional status enables patients to better cope with intensive cancer treatment regimens.

Reference:
1.Cloyd JM, Hyman S, Huwig T, et al. Patient experience and quality of life during neoadjuvant therapy for pancreatic cancer: a systematic review and study protocol. Support Care Cancer. 2021;29(6):3009-3016. doi:10.1007/s00520-020-05813-2.
2.Razdan SN, Patel V, Jewell S, et al. Quality of life among patients after bilateral prophylactic mastectomy: a systematic review of patient-reported outcomes. Qual Life Res. 2016;25(6):1409-1421. doi:10.1007/s11136-015-1181-6.
3.Bowman, Catharine, et al. “The Quality of Life and Psychosocial Implications of Cancer-Related Lower-Extremity Lymphedema: A Systematic Review of the Literature.” Journal of clinical medicine vol. 9,10 3200. 2 Oct. 2020, doi:10.3390/jcm9103200
4. Gosain R, Gupta M, Roy AM, et al. Health-Related Quality of Life (HRQoL) in Neuroendocrine Tumors: A Systematic Review. Cancers (Basel). 2022 Mar 10;14(6):1428. doi: 10.3390/cancers14061428. PMID: 35326587; PMCID: PMC8946839.
5. Devine A, Xiong X, Gottlieb SL, et al. Health-related quality of life in individuals with genital herpes: a systematic review. Health Qual Life Outcomes. 2022 Feb 16;20(1):25. doi: 10.1186/s12955-022-01934-w. PMID: 35172828; PMCID: PMC8848826.

VERSION 2 – REVIEW

REVIEWER	Rahman, M.Mizanur National Institute of Cancer Research and Hospital, Surgical Oncology
REVIEW RETURNED	01-Aug-2022

GENERAL COMMENTS	There is not much work on QOL of breast cancer after NAC. This paper might have some contribution in this field if it is published in the literature.